

CS-30003 · Coding Assignment 1 — Daily Work Schedule

P1: HTTP/1.1 Server & Caching Proxy (Java) · 17 September – 25 October 2026 · 39 days

How the rotation works: the named person is whose turn it is to push that day, in a repeating Aniket → Ahan → Kaustav → Puskar cycle, so each member commits roughly every 4th day and the Git history shows steady, evenly distributed contribution from all four. Aniket leads the cycle because the shared RequestHandler interface and parser are what everyone else builds against. Everyone works on their own module continuously — the rotation governs *who pushes*, not who works. Every member commits under their own name and email; nobody pushes on another member's behalf.

Date	Name	Which part	What to push to GitHub
17 Sep Thu	Aniket Ghosh	TCP Server Core (parsing, status codes, buffering)	Repo init: folder structure, .gitignore, README skeleton, AI-USE.md, and the shared RequestHandler interface so everyone can code in parallel 4 files: README.md, .gitignore, AI-USE.md, src/server/RequestHandler.java
18 Sep Fri	Ahan Bhattacharjee	Thread-Pool Model + Benchmark Harness	Write ConnectionTask (one connection = one unit of work, socket auto-closed) and EchoRequestHandler stub so you can test without the real parser. 2 files: src/server/ConnectionTask.java, src/server/EchoRequestHandler.java
19 Sep Sat	Kaustav Ghosh	Static Serving + Forward Proxy + LRU Cache	Static file serving: read files from a document root, detect MIME type by extension, set Content-Type. 2 files: src/server/StaticFileHandler.java, src/server/MimeTypes.java
20 Sep Sun	Puskar Das	Event-Loop Model + Keep-Alive + Chunked Encoding	Event-loop skeleton: open a Selector, register a non-blocking ServerSocketChannel, empty select() loop. 1 file: src/server/EventLoopServer.java
21 Sep Mon	Aniket Ghosh	TCP Server Core (parsing, status codes, buffering)	Parse the HTTP request line (method, URI, version). Reject anything malformed. 1 file: src/server/HttpRequestParser.java
22 Sep Tue	Ahan Bhattacharjee	Thread-Pool Model + Benchmark Harness	ThreadPoolServer: ServerSocket bind + accept loop that immediately delegates each connection. 1 file: src/server/ThreadPoolServer.java
23 Sep Wed	Kaustav Ghosh	Static Serving + Forward Proxy + LRU Cache	Directory traversal protection: canonicalise paths and reject anything escaping the document root (block ../ attacks). Add a test proving it. 2 files: StaticFileHandler.java (updated), test/TraversalTest.java
24 Sep Thu	Puskar Das	Event-Loop Model + Keep-Alive + Chunked Encoding	Non-blocking accept: handle OP_ACCEPT, register each new client channel for OP_READ. 1 file: EventLoopServer.java (updated)
25 Sep Fri	Aniket Ghosh	TCP Server Core (parsing, status codes, buffering)	Parse headers into a case-insensitive map; handle duplicate and folded headers. 2 files: HttpRequestParser.java (updated), src/server/HttpHeaders.java
26 Sep Sat	Ahan Bhattacharjee	Thread-Pool Model + Benchmark Harness	Configure ThreadPoolExecutor properly: fixed pool size, LinkedBlockingQueue, CallerRunsPolicy. Make pool size a command-line argument. 1 file: ThreadPoolServer.java (updated)
27 Sep Sun	Kaustav Ghosh	Static Serving + Forward Proxy + LRU Cache	Forward proxy mode: accept an absolute-URI request, open an upstream connection, relay the response back. 1 file: src/proxy/ForwardProxy.java
28 Sep Mon	Puskar Das	Event-Loop Model + Keep-Alive + Chunked Encoding	Read-readiness handling: per-connection ByteBuffer state, feed bytes into the shared RequestBuffer. 2 files: EventLoopServer.java (updated), src/server/ConnectionState.java
29 Sep Tue	Aniket Ghosh	TCP Server Core (parsing, status codes, buffering)	Buffering state machine part 1: handle a request split across multiple reads (partial header arrival). 2 files: src/server/RequestBuffer.java, HttpRequestParser.java (updated)
30 Sep Wed	Ahan Bhattacharjee	Thread-Pool Model + Benchmark Harness	Add graceful shutdown (stop accepting, then drain pool) and poolStats() live monitoring. 1 file: ThreadPoolServer.java (updated)
01 Oct Thu	Kaustav Ghosh	Static Serving + Forward Proxy + LRU Cache	LRU cache data structure: bounded-capacity map with least-recently-used eviction. 2 files: src/proxy/LruCache.java, test/LruCacheTest.java
02 Oct Fri	Puskar Das	Event-Loop Model + Keep-Alive + Chunked Encoding	Partial write handling: register OP_WRITE when a response can't be flushed in one go. 1 file: EventLoopServer.java (updated)
03 Oct Sat	Aniket Ghosh	TCP Server Core (parsing, status codes, buffering)	Buffering state machine part 2: handle two or more pipelined requests arriving in a single read. 1 file: RequestBuffer.java (updated)
04 Oct Sun	Ahan Bhattacharjee	Thread-Pool Model + Benchmark Harness	LoadTester: single-request timing path - connect, send GET, read response, stopwatch it. 1 file: src/benchmark/LoadTester.java
05 Oct Mon	Kaustav Ghosh	Static Serving + Forward Proxy + LRU Cache	Parse and honour Cache-Control directives (no-store, no-cache, max-age) when deciding cacheability. 2 files: src/proxy/CachePolicy.java, ForwardProxy.java (updated)
06 Oct Tue	Puskar Das	Event-Loop Model + Keep-Alive + Chunked Encoding	Persistent connections: Connection: keep-alive, reuse the same channel for the next request, idle timeout. 2 files: EventLoopServer.java (updated), src/server/KeepAliveManager.java
07 Oct Wed	Aniket Ghosh	TCP Server Core (parsing, status codes, buffering)	Status codes: 400 (bad request), 404 (not found), 405 (method not allowed), 414 (URI too long), 431 (headers too large). 2 files: src/server/HttpResponse.java, src/server/StatusCode.java
08 Oct Thu	Ahan Bhattacharjee	Thread-Pool Model + Benchmark Harness	LoadTester: concurrency support + p50/p95/p99 percentile and requests/sec calculation. 1 file: LoadTester.java (updated)
09 Oct Fri	Kaustav Ghosh	Static Serving + Forward Proxy + LRU Cache	ETag generation for cached entries and ETag comparison on revalidation. 1 file: src/proxy/ETagGenerator.java
10 Oct Sat	Puskar Das	Event-Loop Model + Keep-Alive + Chunked Encoding	Correct Content-Length handling on responses of known length. 1 file: src/server/ResponseWriter.java
11 Oct Sun	Aniket Ghosh	TCP Server Core (parsing, status codes, buffering)	Implement GET and HEAD properly; add 500 (server error) and 505 (version not supported). 2 files: HttpResponse.java (updated), src/server/RequestDispatcher.java
12 Oct Mon	Ahan Bhattacharjee	Thread-Pool Model + Benchmark Harness	Sweep script: automate runs across concurrency levels (10/50/100/500/1000) and pool sizes, output CSV. 2 files: src/benchmark/SweepRunner.java, scripts/run_sweep.sh
13 Oct Tue	Kaustav Ghosh	Static Serving + Forward Proxy + LRU Cache	If-Modified-Since handling and returning 304 Not Modified instead of a full body. 2 files: CachePolicy.java (updated), src/proxy/ConditionalGet.java
14 Oct Wed	Puskar Das	Event-Loop Model + Keep-Alive + Chunked Encoding	Chunked transfer encoding for responses of unknown length. 2 files: src/server/ChunkedWriter.java, test/ChunkedEncodingTest.java

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15 Oct Thu	Aniket Ghosh	TCP Server Core (parsing, status codes, buffering)	Fuzz tests: feed deliberately fragmented, oversized, and garbage requests; prove the parser survives. 2 files: test/ParserFuzzTest.java, test/PipelineTest.java
16 Oct Fri	Ahan Bhattacharjee	Thread-Pool Model + Benchmark Harness	Run the full thread-pool experiment set; commit raw result CSVs. 2-3 files: results/threadpool_rps.csv, results/threadpool_latency.csv
17 Oct Sat	Kaustav Ghosh	Static Serving + Forward Proxy + LRU Cache	Full conditional-GET flow end-to-end through the proxy; instrument cache hit/miss counters. 2 files: ForwardProxy.java (updated), src/proxy/CacheStats.java
18 Oct Sun	ALL FOUR	Integration checkpoint	Cross-test every module together; both concurrency models must serve the same requests correctly. Integration fixes + test/IntegrationTest.java
19 Oct Mon	Puskar Das	Event-Loop Model + Keep-Alive + Chunked Encoding	Wire the event loop to the real parser and static handler; fix integration issues. 2-3 files: EventLoopServer.java (updated), ResponseWriter.java (updated)
20 Oct Tue	Aniket Ghosh	TCP Server Core (parsing, status codes, buffering)	Fix integration bugs found by the other three; tighten parser edge cases. 2-3 files: parser and buffer fixes (updated)
21 Oct Wed	Ahan Bhattacharjee	Thread-Pool Model + Benchmark Harness	Plotting: RPS-vs-concurrency and latency-percentile plots from the CSVs. 2 files: scripts/plot_results.py, results/plots/*.png
22 Oct Thu	Kaustav Ghosh	Static Serving + Forward Proxy + LRU Cache	Cache experiment: replay a skewed request distribution, measure hit ratio and latency saving vs no-cache. 3 files: src/benchmark/CacheWorkload.java, results/cache_hitratio.csv, results/plots/cache.png
23 Oct Fri	Puskar Das	Event-Loop Model + Keep-Alive + Chunked Encoding	Run the event-loop experiment set with the same harness; commit raw CSVs for the head-to-head comparison. 2 files: results/eventloop_rps.csv, results/eventloop_latency.csv
24 Oct Sat	Aniket Ghosh	TCP Server Core (parsing, status codes, buffering)	Final parser cleanup, inline comments explaining every branch for viva, parser section of report. 2 files: parser files (commented), docs/parser-design.md
25 Oct Sun	ALL FOUR	Final submission prep	Final report, AI-USE.md complete, README with build/run instructions, viva rehearsal. docs/final-report.pdf + README.md (final)